

Functional Derivation of R-2R Ladder DAC Operation

Functional Analysis

January 9, 2026

1 Introduction

This document details the mathematical derivation of the R-2R ladder Digital-to-Analog Converter (DAC). Unlike weighted-resistor architectures, the R-2R ladder utilizes a repeating structure of only two resistor values (R and $2R$) to achieve binary-weighted currents. This derivation analyzes the circuit in **Current Mode**, where the ladder network is terminated at the virtual ground of an Operational Transconductance Amplifier (OTA) or Operational Amplifier.

2 Circuit Analysis

2.1 Equivalent Resistance Property

The fundamental operation of the R-2R ladder relies on the property that the equivalent resistance looking to the right of any node in the ladder is exactly $2R$. We can prove this recursively starting from the Least Significant Bit (LSB).

Consider an N -bit ladder. Let $R_{eq,k}$ be the equivalent resistance looking to the right from node k (where $k = 0$ is the LSB and $k = N - 1$ is the MSB).

1. **Base Case (LSB):** At the LSB end, the ladder is terminated by a $2R$ resistor to ground. The LSB branch itself is also a $2R$ resistor. Looking from the node immediately to the left of the LSB, we see two $2R$ resistors in parallel:

$$R_{eq,0} = 2R \parallel 2R = \frac{2R \times 2R}{2R + 2R} = R \quad (1)$$

2. **Inductive Step:** Moving one stage to the left (towards the MSB), the equivalent resistance $R_{eq,0}$ (which is R) is in series with the horizontal resistor R . This series combination ($R + R = 2R$) is then in parallel with the vertical branch resistor of the current bit ($2R$).

Therefore, for any node k , the equivalent resistance looking to the right is:

$$R_{node_look_right} = (R_{series} + R_{eq,k-1}) \parallel R_{branch} \quad (2)$$

Substituting $R_{series} = R$ and $R_{branch} = 2R$:

$$R_{node} = (R + R) \parallel 2R = 2R \parallel 2R = R \quad (3)$$

Thus, adding the series resistor R restores the resistance seen by the previous node to $2R$. This proves that at every node i along the backbone of the ladder, the equivalent resistance looking towards the LSB is always $2R$.

2.2 Current Division Principle

Since the resistance looking into the right of any node i is $2R$, and the vertical branch resistor at node i is also $2R$, the current $I_{in,i}$ flowing into node i from the left splits equally between the vertical branch (the bit current) and the remainder of the ladder (to the right).

Let I_{ref} be the reference current entering the MSB node.

- **MSB Current (I_{N-1}):** The current splits equally. Half goes to the MSB switch, half goes to the rest of the ladder.

$$I_{N-1} = \frac{I_{ref}}{2} \quad (4)$$

- **Second MSB Current (I_{N-2}):** The remaining $I_{ref}/2$ travels through the series resistor R to the next node, where it splits again.

$$I_{N-2} = \frac{1}{2} \times \left(\frac{I_{ref}}{2} \right) = \frac{I_{ref}}{4} \quad (5)$$

Generalizing for the k -th bit (where $k = 0$ is LSB and $k = N - 1$ is MSB), the branch current I_k is:

$$I_k = \frac{V_{ref}}{R_{total}} \cdot \frac{1}{2^{N-k}} \quad (6)$$

Assuming the network is driven by a voltage reference V_{ref} connected to the ladder input with total characteristic resistance R , the current contribution of the i -th bit (using 1-based indexing for standard formula consistency, $i = 1$ is MSB) is:

$$I_i = \frac{V_{ref}}{2R} \cdot \frac{1}{2^{i-1}} \quad (7)$$

3 Transfer Function Derivation

The total output current I_{out} is the sum of the currents from all branches that are switched to the output line (logic '1'). Branches connected to ground (logic '0') do not contribute to I_{out} .

Let b_i be the digital input bit, where $b_i \in \{0, 1\}$. The total current is:

$$I_{out} = \sum_{i=1}^N b_i \cdot I_i = \sum_{i=1}^N b_i \cdot \frac{V_{ref}}{2^i R} \quad (8)$$

Factorizing common terms:

$$I_{out} = \frac{V_{ref}}{R} \sum_{i=1}^N \frac{b_i}{2^i} \quad (9)$$

In the specific implementation using a 5-Transistor OTA or Op-Amp with feedback resistor R_F , the output current is converted to a voltage. For an inverting configuration:

$$V_{out} = -I_{out} \cdot R_F \quad (10)$$

Substituting the expression for I_{out} :

$$V_{out} = -R_F \left(\frac{V_{ref}}{R} \sum_{i=1}^N \frac{b_i}{2^i} \right) = -V_{ref} \frac{R_F}{R} \left(\frac{b_1}{2^1} + \frac{b_2}{2^2} + \dots + \frac{b_N}{2^N} \right) \quad (11)$$

4 Case Study: 3-Bit DAC

For the specific case of a 3-bit DAC ($N = 3$), the equation expands to:

$$V_{out} = -V_{ref} \frac{R_F}{R} \left(\frac{b_1}{2} + \frac{b_2}{4} + \frac{b_3}{8} \right) \quad (12)$$

Where:

- b_1 is the Most Significant Bit (MSB, D_2).
- b_2 is the middle bit (D_1).
- b_3 is the Least Significant Bit (LSB, D_0).

This confirms that the output voltage is a linear analog representation of the digital input word, scaled by the reference voltage and the gain factor R_F/R .